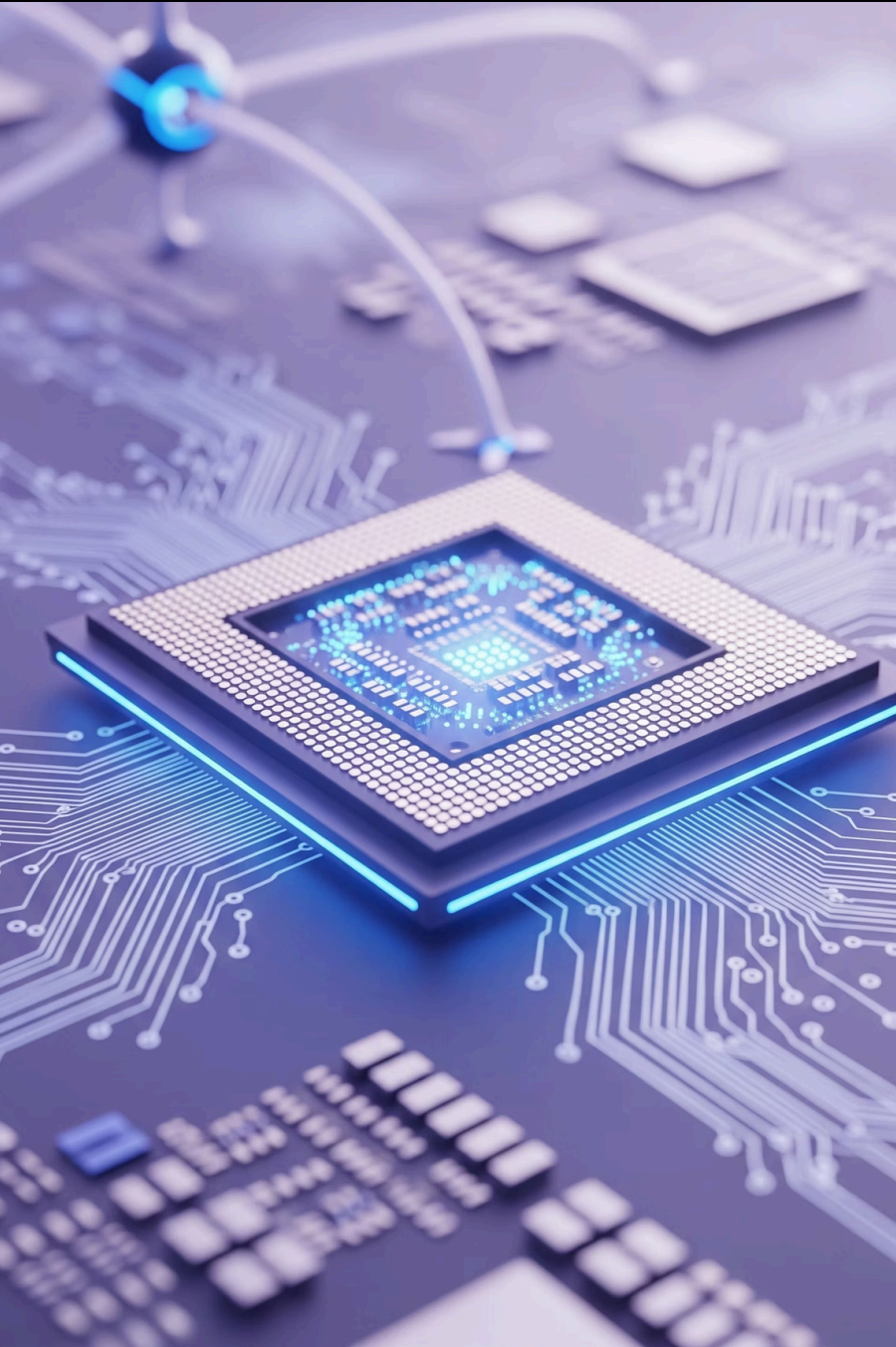




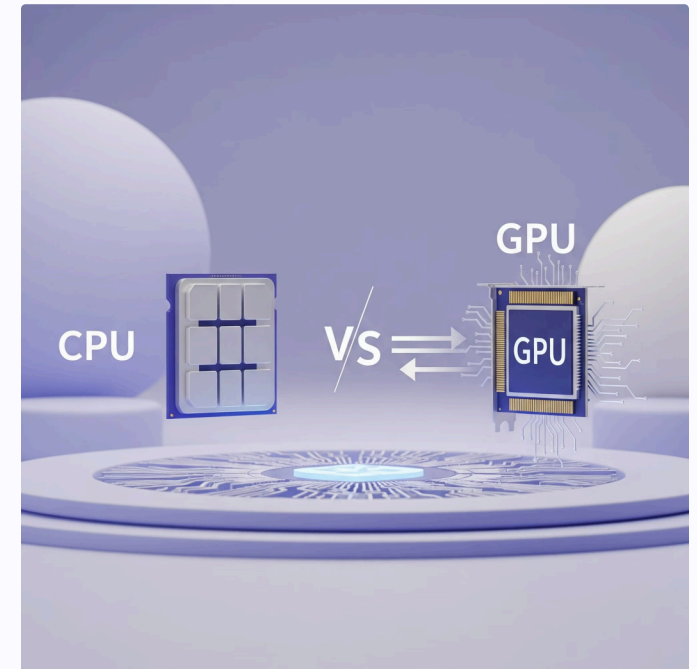
Anatomy of a GPU for AI

A technical exploration of GPU architecture and why it powers modern artificial intelligence infrastructure

Why GPUs Power Modern AI

While CPUs are designed for general-purpose computing, GPUs are specialized parallel accelerators that excel at the **massive matrix multiplications** required by neural networks.

- Process **thousands of operations simultaneously**, making them ideal for AI workloads
- Form the backbone of both deep learning training and inference pipelines
- Essential knowledge for every AI infrastructure engineer



GPU vs CPU Architecture

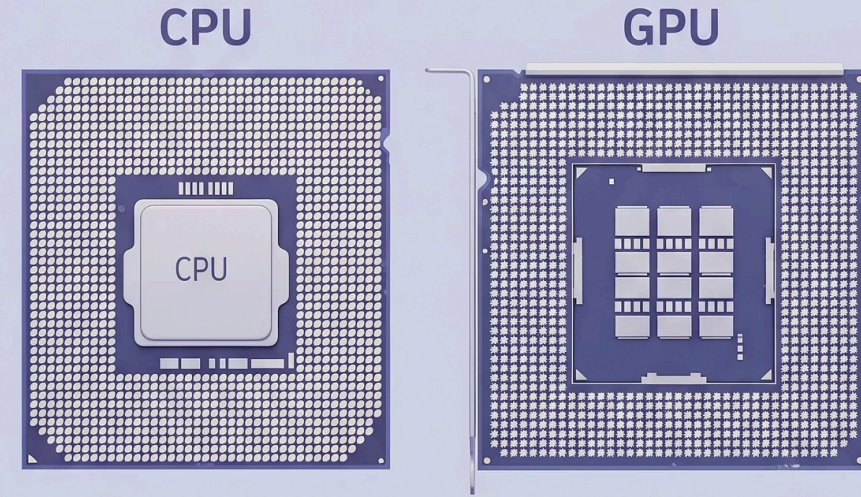
CPU Architecture

- Few powerful cores (4-64)
- Optimized for sequential tasks
- **Latency-focused** design
- Complex control logic and large caches

GPU Architecture

- Thousands of smaller cores
- Designed for massive parallelism
- **Throughput-optimized** design
- Efficiently handles large tensor operations

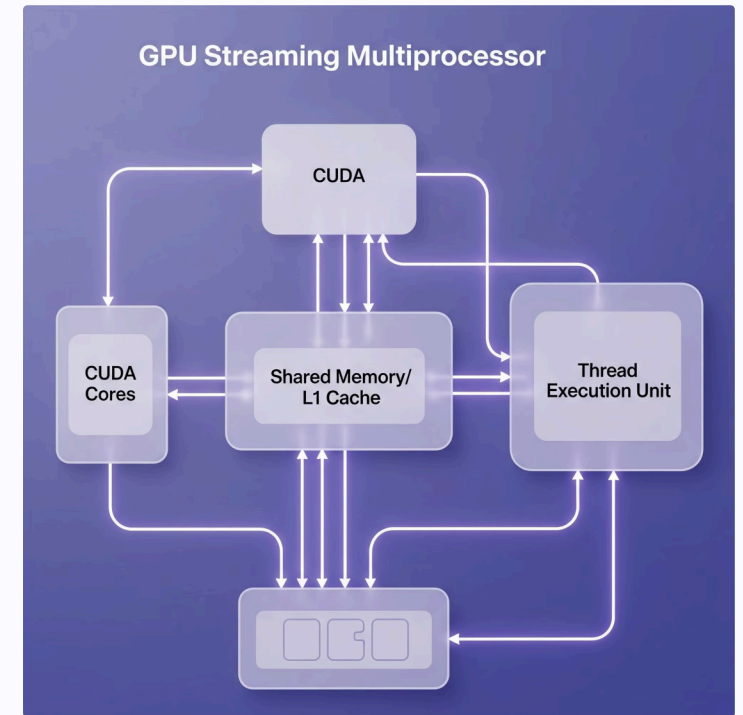
Modern AI systems often utilize **hybrid architectures** leveraging both CPU and GPU strengths, with CPUs handling control flow and GPUs accelerating compute-intensive operations.



GPU Cores & Streaming Multiprocessors

GPU cores are organized into **Streaming Multiprocessors (SMs)** for NVIDIA or **Compute Units (CUs)** for AMD. This architecture enables efficient parallel processing:

- Each SM executes multiple threads concurrently
- **Warps** (groups of 32 threads) execute in lockstep
- Designed for SIMD (Single Instruction, Multiple Data) execution
- Optimized for the matrix/vector operations that power AI



Tensor Cores: AI's Secret Weapon

Introduction

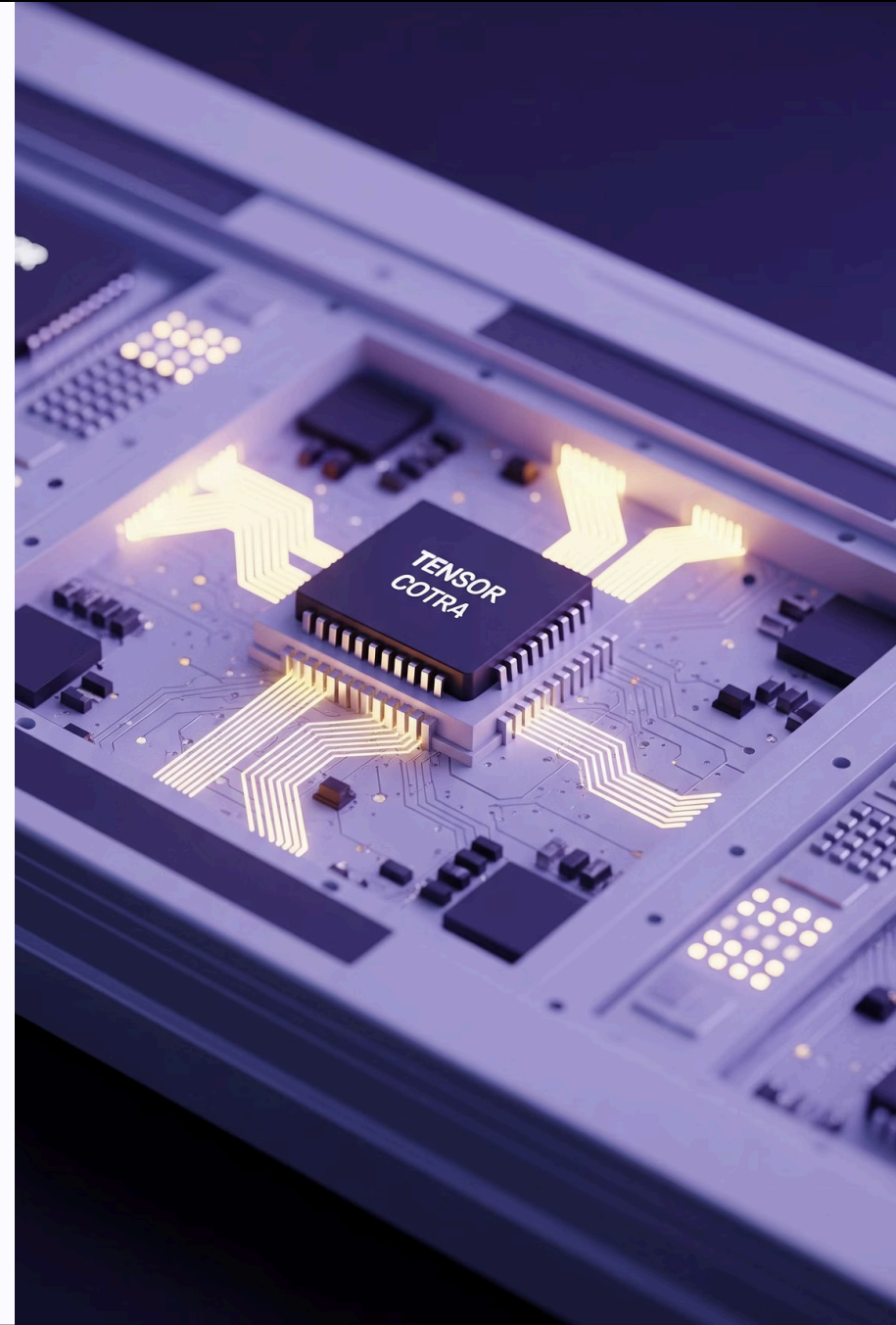
Specialized hardware units for accelerating **matrix multiplication operations**, introduced in NVIDIA's Volta (V100) architecture

Capabilities

Dramatically accelerate FP16, BF16, and INT8 matrix operations common in deep learning

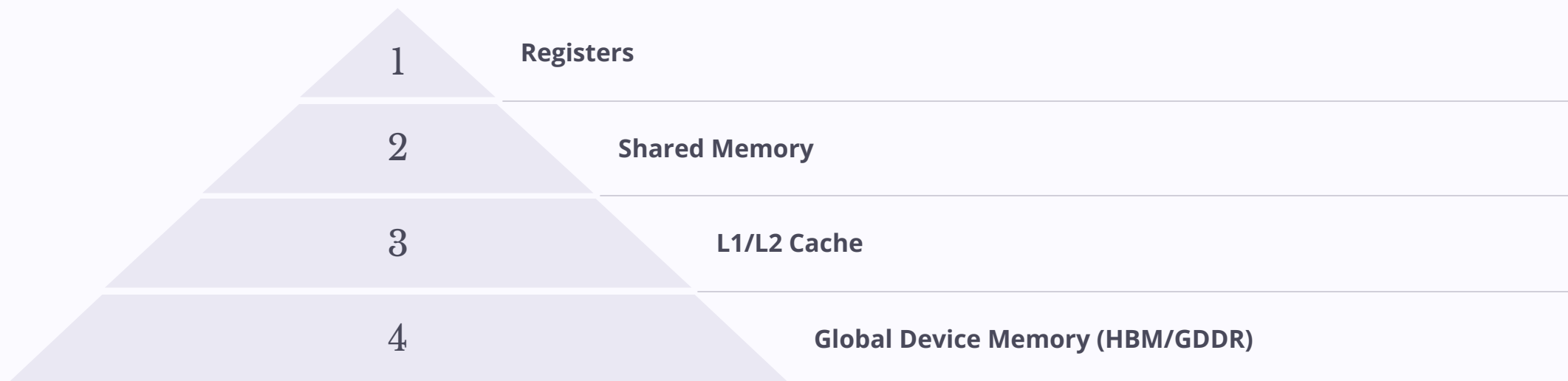
Impact

Deliver **10-100x speedups** for key AI workloads, including LLMs and computer vision



GPU Memory Hierarchy

Memory access patterns critically impact AI workload performance. Engineers must understand the GPU memory hierarchy to optimize their code.



From top to bottom:

- Decreasing speed
- Increasing capacity
- Wider accessibility across threads

Key challenge:

Optimizing AI workloads requires minimizing memory bottlenecks by maximizing data locality and reducing high-latency transfers.

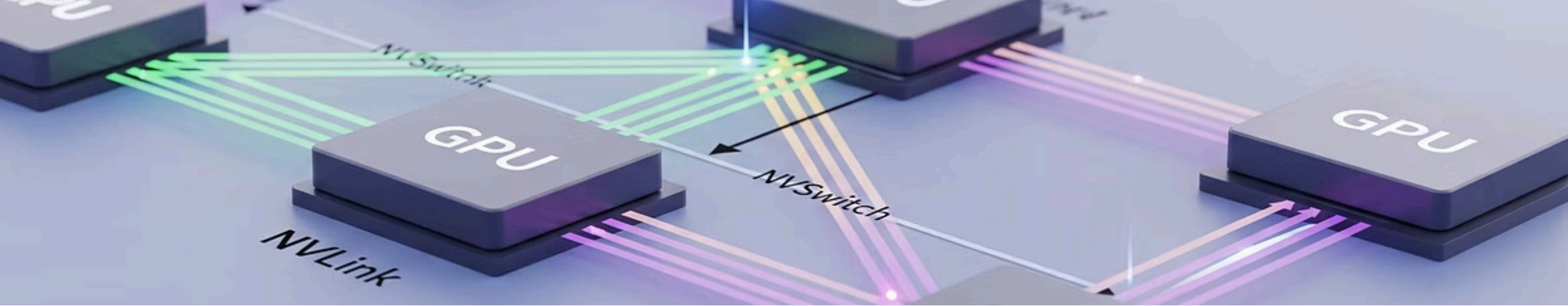
High-Bandwidth Memory (HBM)

Modern AI GPUs utilize **High-Bandwidth Memory** technology to address the memory bottlenecks in large model training and inference:

- Next-gen GPUs use **HBM2e and HBM3** technology
- Provides substantially higher bandwidth than GDDR6
- Enables training of massive Large Language Models
- Example: NVIDIA H100 delivers up to **3 TB/s memory bandwidth**

Memory throughput is often the limiting factor in AI workloads, making HBM a critical advancement.





GPU Interconnects

PCIe

- Standard connection between CPU and GPU
- PCIe 4.0/5.0 offers improved but still limited bandwidth
- Can become a bottleneck in multi-GPU systems

NVLink/NVSwitch

- **Ultra-high bandwidth** GPU-to-GPU connections
- Up to 900 GB/s bidirectional bandwidth (NVLink 4.0)
- Enables efficient multi-GPU communication

Impact on AI

- Critical for **distributed training** of large models
- Reduces communication overhead in model parallelism
- Essential for scaling models like GPT and beyond

GPU Compute Precision

Different AI workloads require different numerical precision. Modern GPUs support multiple precision formats, with **Tensor Cores optimized for mixed-precision operations**.

Format	Bits	Use Case
FP32	32	Standard training
FP16/BF16	16	Efficient training
INT8	8	Inference
INT4	4	Efficient inference



Engineers must balance accuracy vs. speed when selecting precision formats for their AI workloads.

AI Workloads on GPUs

Different AI tasks place different demands on GPU resources:



Training

Large-scale matrix multiplications requiring dense compute and high memory bandwidth. Often distributed across multiple GPUs.



Inference

Lower computational demands but higher sensitivity to latency. Often requires optimization for throughput per watt.



GPU Selection

Depends on model size, batch size, and workload characteristics. Infrastructure engineers must match hardware to specific use cases.

GPUs accelerate diverse AI domains including computer vision, NLP, speech recognition, reinforcement learning, and large language models.

Key Takeaways

GPUs are fundamentally designed for **parallelism and matrix operations**, making them ideal for AI workloads

SMs, Tensor Cores, and memory hierarchy form the backbone of AI acceleration

HBM and interconnects like NVLink enable scaling beyond single-GPU limits

Precision formats allow engineers to balance accuracy and computational efficiency

Next Steps

Mastering GPU architecture is essential for optimizing AI infrastructure. Consider:

- Profiling your workloads to identify bottlenecks
- Experimenting with precision formats
- Optimizing memory access patterns
- Exploring model parallelism for large models

